

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 25: Probability

Topic-wise short notes with solved examples

Probability is a number between 0 and 1 that measures how likely an event is. This chapter covers the probability of an event when all the outcomes are equally likely.

## 1 Basic terms

- A **random experiment** is one whose result cannot be predicted in advance, such as tossing a coin, throwing a die or drawing a card.
- Each possible result is an **outcome**. The set of all possible outcomes is the **sample space**,  $S$ .
- An **event**  $E$  is any set of outcomes. The outcomes in it are the **favourable outcomes**.
- Outcomes are **equally likely** when none is expected to occur more often than another.

## 2 Probability of an event

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} = \frac{n(E)}{n(S)}$$

- $0 \leq P(E) \leq 1$ .
- An **impossible event** has probability 0; a **sure (certain) event** has probability 1.
- The event "not  $E$ " is the **complementary event**  $\bar{E}$ :

$$P(E) + P(\bar{E}) = 1$$

**Example 2.1** A die is thrown once. Find the probability of getting (i) an even number (ii) a multiple of 3 (iii) a number less than 1 (iv) a number not greater than 6.

**SOLUTION**

$S = \{1, 2, 3, 4, 5, 6\}$ , so  $n(S) = 6$

(i)  $\{2, 4, 6\}$ :  $P = \frac{3}{6} = \frac{1}{2}$  (ii)  $\{3, 6\}$ :  $P = \frac{2}{6} = \frac{1}{3}$

(iii) No such number:  $P = 0$  (iv) All six numbers:  $P = 1$

(i)  $\frac{1}{2}$  (ii)  $\frac{1}{3}$  (iii) 0 (iv) 1

### 3 Coins and dice

| Experiment  | Number of outcomes | Sample space                           |
|-------------|--------------------|----------------------------------------|
| One coin    | 2                  | H, T                                   |
| Two coins   | 4                  | HH, HT, TH, TT                         |
| Three coins | 8                  | HHH, HHT, HTH, THH, HTT, THT, TTH, TTT |
| One die     | 6                  | 1, 2, 3, 4, 5, 6                       |
| Two dice    | 36                 | Ordered pairs (1, 1) to (6, 6)         |

**Example 3.1** Three coins are tossed together. Find the probability of getting (i) exactly two heads (ii) at least one tail.

**SOLUTION**

$$n(S) = 8$$

$$(i) \text{ HHT, HTH, THH: } P = \frac{3}{8}$$

$$(ii) \text{ "At least one tail" is the complement of "no tail" (HHH): } P = 1 - \frac{1}{8}$$

$$(i) \frac{3}{8} \quad (ii) \frac{7}{8}$$

**Example 3.2** Two dice are thrown together. Find the probability that (i) the sum is 7 (ii) the two numbers are equal (iii) the sum is more than 10.

**SOLUTION**

$$n(S) = 36$$

$$(i) (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1): P = \frac{6}{36} = \frac{1}{6}$$

$$(ii) (1, 1), (2, 2), \dots, (6, 6): P = \frac{6}{36} = \frac{1}{6}$$

$$(iii) (5, 6), (6, 5), (6, 6): P = \frac{3}{36} = \frac{1}{12}$$

$$(i) \frac{1}{6} \quad (ii) \frac{1}{6} \quad (iii) \frac{1}{12}$$

### 4 Playing cards

- A pack has **52 cards**: 4 suits of 13 cards each.
- Spades and clubs are **black** (26 cards); hearts and diamonds are **red** (26 cards).
- Each suit has an ace, the numbers 2 to 10, a jack, a queen and a king.
- **Face cards**: jack, queen and king. There are 12 face cards, 6 red and 6 black.

**Example 4.1** A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is (i) an ace (ii) a black king (iii) a face card (iv) neither a heart nor a king.

**SOLUTION**

(i) 4 aces:  $P = \frac{4}{52} = \frac{1}{13}$  (ii) 2 black kings:  $P = \frac{2}{52} = \frac{1}{26}$

(iii) 12 face cards:  $P = \frac{12}{52} = \frac{3}{13}$

(iv) Hearts or kings: 13 hearts + 3 other kings = 16 cards, so  $52 - 16 = 36$  cards:  $P = \frac{36}{52} = \frac{9}{13}$

(i)  $\frac{1}{13}$  (ii)  $\frac{1}{26}$  (iii)  $\frac{3}{13}$  (iv)  $\frac{9}{13}$

## 5 Balls, numbered cards and letters

**Example 5.1** A bag contains 4 red, 6 blue and 5 green balls. One ball is drawn at random. Find the probability that it is (i) blue (ii) not green (iii) red or green.

**SOLUTION**

Total = 15 balls

(i)  $P = \frac{6}{15} = \frac{2}{5}$  (ii)  $P = \frac{10}{15} = \frac{2}{3}$  (iii)  $P = \frac{9}{15} = \frac{3}{5}$

(i)  $\frac{2}{5}$  (ii)  $\frac{2}{3}$  (iii)  $\frac{3}{5}$

**Example 5.2** Cards numbered 1 to 25 are placed in a box and one is drawn at random. Find the probability that the number is (i) prime (ii) a perfect square (iii) divisible by both 2 and 3.

**SOLUTION**

(i) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, which is 9 numbers:  $P = \frac{9}{25}$

(ii) 1, 4, 9, 16, 25:  $P = \frac{5}{25} = \frac{1}{5}$

(iii) Multiples of 6: 6, 12, 18, 24:  $P = \frac{4}{25}$

(i)  $\frac{9}{25}$  (ii)  $\frac{1}{5}$  (iii)  $\frac{4}{25}$

**Example 5.3** A letter is chosen at random from the word PROBABILITY. Find the probability that it is (i) a vowel (ii) the letter B.

**SOLUTION**

The word has 11 letters: P, R, O, B, A, B, I, L, I, T, Y.

(i) Vowels: O, A, I, I, which is 4 letters:  $P = \frac{4}{11}$

(ii) B appears twice:  $P = \frac{2}{11}$

(i)  $\frac{4}{11}$  (ii)  $\frac{2}{11}$

## 6 Finding an unknown number

**Example 6.1** A bag contains 12 balls, of which  $x$  are white. (i) Write the probability of drawing a white ball. (ii) If 6 more white balls are added, the probability of drawing a white ball doubles. Find  $x$ .

**SOLUTION**

(i)  $P = \frac{x}{12}$

(ii) Now there are  $x + 6$  white balls out of 18:  $\frac{x + 6}{18} = 2 \times \frac{x}{12}$

$\frac{x + 6}{18} = \frac{x}{6}$ , so  $x + 6 = 3x$

$x = 3$

## 7 Common mistakes

- Writing a probability greater than 1.
- Treating (3, 4) and (4, 3) as the same outcome for two dice.
- Counting 1 as a prime number.
- Counting the king of hearts twice in "a heart or a king".
- Leaving the answer unsimplified.